



Network Implementation Report

1. Introduction

My role as part of the Network Configuration Team in this project involved the implementation of an enterprise network, using Cisco Packet Tracer. We chose a hierarchical leaf-spine architecture; which seemed suited to create a scalable, secure, and manageable network capable of supporting multiple departments while providing centralized network services such as DHCP and DNS.

The network was designed to provide logical segmentation through Virtual Local Area Networks (VLANs), centralized routing through Layer 3 switches, and dedicated infrastructure for internal services, as well as public-facing resources.

2. Network Topology

The network follows a hierarchical design consisting of a core (spine) layer, distribution (leaf) layer, and access layer.

The topology is composed of:

- * Two Core (Spine) Layer 3 switches
- * Three Leaf Layer 3 switches
- * Multiple access switches
- * DHCP server
- * DNS server
- * Internal servers
- * DMZ servers
- * End-user workstations

The three leaf switches serve different purposes:

- * Leaf 1 connects all departmental access switches.
- * Leaf 2 connects the internal server infrastructure.
- * Leaf 3 connects the DMZ network hosting public-facing services.

Each department is connected to the network through its own dedicated access switch, providing clear separation between departments and simplifying management.

3. VLAN Configuration

To improve security, reduce broadcast traffic, and logically separate departments, multiple VLANs were created:

VLAN	Purpose
VLAN 10	IT Department

VLAN 20	Production Department
VLAN 30	Management Department
VLAN 40	Support Department 1
VLAN 50	Support Department 2
VLAN 60	Study Department
VLAN 70	Internal Servers
VLAN 80	DMZ
VLAN 90	iSCi Storage
VLAN 99	Guest Access

Each access switch was configured so that all user ports belonged to the appropriate departmental VLAN, while all uplinks towards Leaf 1 operated as IEEE 802.1Q trunk links. Unused interfaces were assigned to VLAN 999 and administratively shut down to improve network security.

4. Layer 3 Routing

Inter-VLAN routing was performed directly on the Layer 3 infrastructure. Switch Virtual Interfaces (SVIs) were created for every VLAN.

Each SVI was assigned the first usable IP address within its subnet and served as the default gateway for devices in that VLAN.

Each Layer 3 switch was configured with the ip routing command, enabling communication between all VLANs while maintaining network segmentation.

5. VLAN Trunking

Trunk links were configured between:

- * Core and Leaf switches
- * Leaf switches and Access switches

Each trunk carried only the VLANs required by that section of the network.

The native VLAN was configured as VLAN 99 to reduce the risk of VLAN hopping attacks. Allowed VLAN lists were configured to prevent unnecessary VLAN propagation throughout the network.

6. IP Addressing

Each VLAN received its own subnet and default gateway.

7. DHCP Configuration

A centralized DHCP server was deployed within VLAN 70.

Separate DHCP pools were created for every user VLAN.

Each DHCP pool contained:

- * Network address
- * Subnet mask
- * Default gateway
- * DNS server address
- * Starting IP address
- * Maximum number of users

Because the DHCP server resided in a different VLAN, DHCP relay was configured using the ip helper-address command on every user VLAN SVI.

This allowed DHCP broadcast requests to be forwarded to the centralized DHCP server.

Client devices were successfully configured to obtain IP addresses automatically.

8. DNS Configuration

A dedicated DNS server was deployed within the server VLAN.

The DNS service was enabled and configured with resource records for internal services.

Examples include:

- * Web server
- * FTP server
- * DHCP server

Client computers received the DNS server address automatically through DHCP, allowing hostnames to be resolved without requiring users to remember IP addresses.

Successful hostname resolution confirmed that the DNS service operated correctly.

9. Server Infrastructure

The internal server network was isolated within VLAN 70.

Services included:

- * DHCP
- * DNS
- * File services

Public-facing services were placed inside the DMZ (VLAN 80).

Separating public servers from the internal network significantly improved security by limiting direct access to internal resources.

10. Switch Configuration

Each access switch was configured with:

- * Access ports assigned to the correct VLAN
- * Trunk uplinks toward Leaf 1
- * Remote management through VLAN 99
- * Shutdown of unused interfaces

Leaf switches aggregated traffic from the access layer and forwarded VLAN traffic toward the core.

The Core Layer 3 switch provided:

- * Inter-VLAN routing
- * DHCP relay
- * Default gateways
- * Routing between internal networks

11. Testing and Verification

After implementation, the following tests were performed:

- * Verification of VLAN membership
- * Verification of trunk links
- * DHCP address assignment
- * DNS hostname resolution
- * Inter-VLAN communication

- * Connectivity to internal servers
- * Connectivity to DMZ resources
- * Verification of switch management through VLAN 99

All devices successfully obtained IP addresses from the centralized DHCP server.

Hostnames were resolved correctly through the DNS server.

Communication between VLANs functioned as expected after routing was enabled.

12. Challenges Encountered

Several configuration issues were encountered during implementation, requiring troubleshooting before the network became fully operational.

12.1 IP Routing

Initially, devices in different VLANs were unable to communicate because Layer 3 routing had not been fully configured.

The issue was resolved by enabling ip routing on the Layer 3 switch and verifying that all Switch Virtual Interfaces were active.

12.2 VLAN Configuration

Some devices initially received incorrect network connectivity due to interfaces being assigned to the wrong VLAN.

This required reviewing VLAN assignments and verifying switchport configurations on multiple access switches.

12.3 Switch Virtual Interfaces (SVIs)

Several SVIs remained in a down state because no active ports existed within their associated VLANs.

This issue was resolved by ensuring that each VLAN had at least one active access port or trunk connection.

Correct IP addressing and subnet masks were also verified.

12.4 Trunk Configuration

One of the most significant challenges involved configuring trunk links between switches.

Some trunk ports initially failed to carry all required VLANs because of mismatched trunk configurations and incomplete allowed VLAN lists.

After reviewing trunk status using Cisco IOS verification commands and correcting the native VLAN configuration, all VLAN traffic passed successfully across the network.

12.5 Spanning Tree Protocol

During the implementation, several end devices experienced delayed connectivity after being connected to the network. Although the VLANs and trunk links were

configured correctly, the access ports remained in the default Spanning Tree Protocol (STP) listening and learning states before transitioning to the forwarding state. This prevented client devices from immediately communicating with the network and, in some cases, delayed DHCP address assignment.

The issue was resolved by enabling PortFast on all downlink interfaces connecting the access switches to end-user devices

12.6 DHCP Relay

Although the DHCP server was correctly configured, clients in other VLANs initially failed to receive addresses.

The issue was traced to missing ip helper-address statements on several VLAN interfaces.

After configuring DHCP relay, clients across all departmental VLANs successfully obtained addresses from the centralized DHCP server.

13. Conclusion

The implementation successfully demonstrated the deployment of an enterprise network using a hierarchical leaf-spine architecture with Cisco Catalyst 3650 Layer 3 switches. The network incorporated VLAN segmentation, centralized inter-VLAN routing, DHCP, DNS, server isolation, a dedicated DMZ, and remote management capabilities.

Although several challenges were encountered during the configuration of routing, VLANs, trunk links, Switch Virtual Interfaces, DHCP relay, and Spanning Tree Protocol, each issue was systematically identified and resolved through testing and troubleshooting.

The completed network provides a scalable, secure, and manageable infrastructure suitable for supporting multiple departments while following industry best practices for enterprise campus network design.